

BRADLEY DIRKS

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POSITIONS

Stony Brook University

Simons Instructor

2023-2027

Stony Brook, NY

NSF Mathematical Sciences Postdoctoral Research Fellow

2023-2027

Stony Brook University

Stony Brook, NY

Institute for Advanced Study

2024-2025

Member, School of Mathematics

Princeton, NJ

EDUCATION

University of Michigan

2018-2023

PhD in Mathematics. Advisor: Mircea Mustață

Ann Arbor, MI

- Thesis: Using Mixed Hodge Modules to Study Singularities

University of California

2014-2018

BS and MA in Mathematics

Los Angeles, CA

RESEARCH INTERESTS

Algebraic Geometry, Singularities, $\mathcal{D}$ -modules, Hodge Theory

PUBLICATIONS

1. **Restrictions of Hodge modules using generalized V -filtrations**
with Qianyu Chen and Sebastián Olano. *Adv. Math.* **499** (2026)
2. **Fourier transform and Radon transform for mixed Hodge modules**
To appear in *Ann. Inst. Fourier*
3. **Some applications of Microlocalization for LCI subvarieties**
Math. Ann. **392** (2025)
4. **Verdier specialization and restrictions of Hodge modules**
with Qianyu Chen and Morihiko Saito. *Math. Z.* **310** (2025)
5. **The minimal exponent of cones over smooth complete intersection projective varieties**
with Qianyu Chen and Mircea Mustață. *Rev. Roumaine Math. Pures Appl.* **70** (2025)
6. **The minimal exponent and k -rationality for locally complete intersections**
with Qianyu Chen and Mircea Mustață. *J. Ec. polytech. Math.* **11** (2024)
7. **V -filtrations and minimal exponents of locally complete intersection singularities**
with Qianyu Chen, Mircea Mustață and Sebastián Olano. *J. Reine Angew. Math.* **811** (2024)
8. **An introduction to V -filtrations**
with Qianyu Chen and Mircea Mustață. *Handbook of Geometry and Topology of Singularities, Volume VII* (2025)
9. **On V -filtration, Hodge filtration and Fourier Transform**
with Qianyu Chen. *Selecta Math. (N.S.)* **29** (2023)
10. **Minimal exponents of hyperplane sections: a conjecture of Teissier**
with Mircea Mustață. *J. Eur. Math. Soc.* **25** (2023)
11. **Upper bounds for roots of B-functions, following Kashiwara and Lichtin**
with Mircea Mustață. *Publ. Res. Inst. Math. Sci.* **58** (2022)
12. **The Hilbert series of Hodge ideals of hyperplane arrangements**
with Mircea Mustață. *J. Singul.* **20** (2020)

PREPRINTS

1. **Relative Du Bois complexes through mixed Hodge modules**
Preprint.
2. **Microlocal Bernstein-Sato polynomials on singular ambient varieties**
with Michael Perlman. Preprint.
3. **Filtrations on local cohomology and Injectivity Theorems**
with Qianyu Chen and Sebastián Olano. Preprint.
4. **F -nilpotence and Hodge filtrations beyond isolated singularities**
with Jakub Witaszek. Preprint.
5. **Simple characterizations and finite descent properties of some Hodge theoretic singularity invariants**
with Sebastián Olano and Debaditya Raychaudhury. Preprint.
6. **Multiplier modules, V -filtrations and Bernstein-Sato polynomials on singular ambient varieties**
Preprint.
7. **Hodge theory of secant varieties**
with Qianyu Chen, Sebastián Olano and Debaditya Raychaudhury. Preprint.
8. **Partial cohomologically complete intersections via Hodge theory**
with Qianyu Chen and Sebastián Olano. Preprint.
9. **Hirzebruch-Milnor classes of local complete intersections, minimal exponent, and applications to higher singularities**
with Laurențiu Maxim and Sebastián Olano. Preprint.
10. **A Hodge theoretic generalization of $\mathbb{Q}$ -homology manifolds II: Local Complete Intersections**
with Sebastián Olano and Debaditya Raychaudhury. Preprint.
11. **A Hodge theoretic generalization of $\mathbb{Q}$ -homology manifolds I: General Case**
with Sebastián Olano and Debaditya Raychaudhury. Preprint.

AWARDS

US Junior Oberwolfach Fellow	MFO, 2025
NSF Mathematical Sciences Postdoctoral Research Fellowship	National Science Foundation, 2023
Wirt and Mary Cornwell Prize	University of Michigan Mathematics Department, 2023
Rackham Predoctoral Fellowship	University of Michigan, Summer 2022-Winter 2023
Paul Daus Memorial Award	UCLA, 2018
Departmental Honors	UCLA Math Department, 2018
Summa Cum Laude	UCLA, 2018

INVITED TALKS

BIRS Workshop on Singularities, D-modules and Connections to Physics	Sep. 2025
<i>Hodge theory of the constant sheaf</i>	BIRS
Singularities in the Midwest, IX	April 2025
<i>Hodge theoretic properties of LCI singularities</i>	University of Wisconsin, Madison
AMS 2025 Spring Sectional Meeting	March 2025
<i>Singularities of Local Complete Intersections via Hodge Theory</i>	University of Kansas
JHU Algebraic Geometry Seminar	October 2024
<i>Minimal Exponent for LCI subvarieties</i>	Johns Hopkins University
Princeton University Algebraic Geometry Seminar	September 2024
<i>Minimal Exponent for LCI subvarieties</i>	Princeton University
Birational Geometry Seminar 2024	May 2024
<i>Recent results on minimal exponent for LCI subvarieties</i>	Online
CUNY Commutative Algebra and Algebraic Geometry Seminar	March 2024
<i>The minimal exponent for LCI subvarieties</i>	Graduate Center of CUNY
University of Utah Algebraic Geometry Seminar	February 2024
<i>The minimal exponent for LCI subvarieties</i>	University of Utah

Columbia University Algebraic Geometry Seminar <i>The minimal exponent for LCI subvarieties</i>	January 2024 Columbia University
Winter School on New Applications of Mixed Hodge Modules <i>V-filtration and Hodge filtration in higher codimension</i>	January 2024 Simons Center for Geometry and Physics
University of Toronto Algebraic Geometry Seminar <i>The minimal exponent for LCI subvarieties</i>	November 2023 University of Toronto
Harvard/MIT Algebraic Geometry Seminar <i>The minimal exponent for LCI subvarieties</i>	October 2023 Harvard University
Birational Geometry Seminar 2023 <i>Higher Du Bois and higher rational singularities for LCI varieties</i>	May 2023 Online
MAGGC <i>V-filtrations of $\mathcal{D}$-modules</i>	August 2022 UIC
Algebraic Geometry and Singularities Learning Workshop & Conference <i>Comparing V-filtration of an ideal with that of a general linear combination</i>	June 2022 UW Seattle
Algebraic Geometry Seminar <i>The Structure of Monodromic Mixed Hodge modules</i>	February 2022 Stony Brook University
DOCAS Seminar <i>Understanding the roots of b-functions</i>	August 2021 Online
UConn Algebra Seminar <i>Minimal Exponents and a Conjecture of Teissier</i>	April 2021 Online
Topology and Singularities Seminar in Madison <i>Minimal Exponents and a Conjecture of Teissier</i>	March 2021 Online

ORGANIZING

Topical Workshop on Singularities and Syzygies of Algebraic Varieties <i>ICERM, Brown University</i>	August 2027 Co-organized
2026 Spring Eastern Sectional Meeting <i>Boston College</i>	March 2026 w/ Sebastián Olano, Debaditya Raychaudhury and Wanchun Shen
Learning Seminar on mixed Hodge modules <i>Princeton University</i>	Fall 2024 w/ Bhargav Bhatt, Sung Gi Park and Jakub Witaszek
AIM Workshop “Higher Du Bois and higher rational singularities” <i>American Institute of Mathematics. See website</i>	October 28 - November 1, 2024 w/ Radu Laza
Winter School on New Applications of Mixed Hodge Modules <i>Simons Center for Geometry and Physics. See website</i>	January 2024 w/ Christian Schnell
Student Algebraic Geometry seminar <i>University of Michigan</i>	Winter 2021-2023 w/ Devlin Mallory in Winter 2021, w/ Saket Shah 2022-2023
$\mathcal{D}$-modules and Representation Theory Minicourse <i>University of Michigan</i>	Summer 2022
Mixed Hodge Theory Minicourse <i>University of Michigan</i>	Summer 2020 w/ James Hotchkiss
Variations of Hodge Structure Reading Group <i>University of Michigan</i>	Winter 2020 w/ James Hotchkiss

ADVISING

Master’s Thesis Advising <i>Stony Brook University</i> • Zirui Liu – Complex Analytic Geometry	2026–2027 Stony Brook, NY
Master’s Research Advising <i>Stony Brook University</i> • Andrew Lin – Bernstein–Sato polynomials in positive characteristic	2026–2027 Stony Brook, NY
Master’s Reading Course <i>Stony Brook University</i> • Zhiqing Chen – Coutinho’s <i>A Primer of Algebraic $\mathcal{D}$-modules</i> and Hotta-Takeuchi-Tanisaki <i>$\mathcal{D}$-modules, Perverse Sheaves and Representation Theory</i>	Summer 2026 Stony Brook, NY

TEACHING EXPERIENCE

Instructor

Stony Brook University

- MAT544: Commutative and Homological Algebra

Fall 2026

Stony Brook, NY

Instructor

Stony Brook University

- MAT310: Linear Algebra

Spring 2024

Stony Brook, NY

Graduate Student Instructor

University of Michigan

- Math 105 (Precalculus), Math 115 (Calculus I), Math 116 (Calculus II)

October 2018-Winter 2022

Ann Arbor, MI

Docent

LA Math Circle (now ORMC)

2016 - 2018

Los Angeles, CA